

Minimal Additive Codes

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Why are minimal codes interesting ?

- Minimal codewords were first studied by Massey in 1993. He proposed an application to Secret Sharing Schemes: supports of minimal codewords of some code were used to define an access structure to a secret.



Massey “Minimal codewords and secret sharing”, Proceedings of the 6th joint Swedish-Russian international workshop on information theory, 1993



Chatterjee, Mesnager, Mishra, Maji “Structure and Construction of Two-Dimensional Minimal Linear Codes over the rings $\mathbb{Z}_p^n$ with Applications to Secret Sharing”, 2026

- They have application in additive and extremal combinatorics.



Bishnoi, D'haeseleer, Gijswijt, Potukuchi “Blocking sets, minimal codes and trifferent codes”, 2023



Kurz “Trifferent codes with small lengths”, 2023

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① Additive codes

② Geometric point of view

③ Minimality for additive codes

Framework : Hamming metric

- A **code** is a subset of $\mathbf{F}^n$, its elements are called **codewords**
- The **support** of a codeword $c = (c_1, \dots, c_n) \in \mathbf{F}^n$ is the set

$$\text{supp}(c) := \{i \in \{1, \dots, n\} : c_i \neq 0\}$$

- The **Hamming weight** of c is $\text{wt}(c) = |\text{supp}(c)|$
- The **Hamming distance** is defined as $d(c, c') = \text{wt}(c - c')$
- The **minimum Hamming distance** of a code $\mathcal{C}$ is defined as

$$d := \min\{d(c, c') : c, c' \in \mathcal{C}, c \neq c'\}$$

Linear codes

- Ambient space: $\mathbf{F}_q^n$
- Code: $\mathcal{C}$ is an $\mathbf{F}_q$ -subspace of $\mathbf{F}_q^n$
- Dimension: $k = \dim_{\mathbf{F}_q}(\mathcal{C}) = \log_q |\mathcal{C}|$
- Minimum distance: $d = \min\{\text{wt}(c) : c \in \mathcal{C} \setminus \{0\}\}$
- Notation: $[n, k, d]_q$ code
- Generator matrix: $G \in \mathbf{F}_q^{k \times n}$

$$\mathcal{C} = \{xG : x \in \mathbf{F}_q^k\}$$

- Nondegeneracy: $\nexists i \in [n]$ s.t. $\forall c \in \mathcal{C}, c_i = 0$
- Equivalence: $\mathcal{C}, \mathcal{D}$ are **equivalent** if there exists a map $\phi \in (\mathbf{F}_q^*)^n \rtimes \mathcal{S}_n$ such that

$$\phi(\mathcal{C}) = \mathcal{D}$$

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Example

Consider $\mathbf{F}_4 = \mathbf{F}_2(\omega)$, with ω a root of $x^2 + x + 1 \in \mathbf{F}_2[x]$.

Let $\mathcal{C}$ be the $[5, 2/2]_2^2$ additive code defined by the following generator matrix

$$G = \begin{pmatrix} 1 & 1 & \omega & 0 & 0 \\ 1 & \omega & 0 & 1 & \omega^2 \end{pmatrix}.$$

We have that

$$\mathcal{C} = \{xG : x \in \mathbf{F}_2^2\} = \{(0, 0, 0, 0, 0), (1, 1, \omega, 0, 0), (1, \omega, 0, 1, \omega^2), (0, \omega^2, \omega, 1, \omega^2)\}.$$

The minimum distance of $\mathcal{C}$ is $d = 3$.

- ① Additive codes
- ② Geometric point of view
- ③ Minimality for additive codes

Projective systems

Definition

A multiset $\mathcal{P}$ of n points of $\text{PG}(k-1, q)$ is an $[n, k, d]_q$ projective system if $\mathcal{P}$ is not contained in any hyperplane $H \subseteq \text{PG}(k-1, q)$ and

$$d = n - \max_{\substack{H \subseteq \text{PG}(k-1, q) \\ H \text{ hyperplane}}} |\mathcal{P} \cap H|.$$

- **Equivalence:** $\mathcal{P}$ and $\mathcal{Q}$ are **equivalent** if there exists $\psi \in \mathbf{PGL}_k(\mathbf{F}_q)$ such that $\psi(\mathcal{P}) = \mathcal{Q}$.

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Definition

A multiset $\chi = \{\pi_1, \dots, \pi_n\}$ of n **subspaces** of $\text{PG}(k-1, q)$ with $\dim(\pi_i) \leq h-1$ is an h - $[n, k, d]_q$ projective system if the π_i are not all contained in the same hyperplane and

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- **Equivalence:** χ and γ are **equivalent** if there exists $\psi \in \text{PGL}_k(\mathbf{F}_q)$ such that $\psi(\chi) = \gamma$.

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Theorem

There is a one-to-one correspondence between (equivalence classes of) nondegenerate $[n, k, d]_q$ codes and (equivalence classes of) projective $[n, k, d]_q$ systems.



Tsfasman and Vladut. "Algebraic-geometric codes".

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Theorem (Ball, Lavrauw, Popatia)

There is a one-to-one correspondence between (equivalence classes of) nondegenerate $[n, k/h, d]_q^h$ additive codes and (equivalence classes of) projective h - $[n, k, d]_q$ systems.



Ball, Lavrauw, Popatia. "Griesmer type bounds for additive codes over finite fields", 2024

Nondegenerate additive code $\rightarrow h$ -projective system

Let $\mathcal{C}$ be a nondegenerate $[n, k/h, d]_q^h$ code and let $G \in \mathcal{M}_{k,n}(\mathbf{F}_{q^h})$ be a generator matrix for $\mathcal{C}$:

$$G = (\textcolor{red}{g}_1 \mid \textcolor{green}{g}_2 \mid \cdots \mid \textcolor{blue}{g}_n),$$

where $g_i = (g_{i,1}, \dots, g_{i,k})^\top \in \mathbf{F}_{q^h}^k$. Let $\beta = (\alpha_1, \dots, \alpha_h)$ be a basis of $\mathbf{F}_{q^h}/\mathbf{F}_q$ and write

$$g_{j,i} := \sum_{t=1}^h b_{j,i}^t \alpha_t,$$

where $b_{i,j}^t \in \mathbf{F}_q$ for every i, j, t .

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This yields a matrix $\tilde{G} \in \mathcal{M}_{k,nh}(\mathbb{F}_q)$:

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Let define $\pi_i := \text{PG}(\text{colsp}_{\mathbb{F}_q}(\tilde{G}_i)) \subseteq \text{PG}(k-1, q)$ of dimension at most $h-1$.

Nondegenerate additive code $\rightarrow$ h -projective system

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A codeword $c := xG \in \mathcal{C}$ with $x \in \mathbb{F}_q^k \setminus \{0\}$, geometrically c corresponds to the hyperplane

$$\langle x \rangle^\perp \subseteq \text{PG}(k-1, q).$$

The weight of c can be obtained as

$$\begin{aligned} \text{wt}(c) &= \text{wt}(xG) \\ &= n - |\{i \in \{1, \dots, n\} : \pi_i \subseteq \langle x \rangle^\perp\}|. \end{aligned}$$

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Therefore, the minimum distance of the code $\mathcal{C}$ is

$$d = n - \max_{\substack{H \subseteq \text{PG}(k-1, q), \\ \bar{H} \text{ hyperplane}}} |\{i \in \{1, \dots, n\} : \pi_i \subseteq H\}|.$$

Therefore, χ is an h - $[n, k, d]_q$ projective system.

Back to the previous example

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The weight of the codeword $c := (1 \ 0) G = (1 \ 1 \ \omega \ 0 \ 0)$ is

$$\text{wt}(c) = 5 - |\{i : \pi_i \subseteq \langle [1 : 0] \rangle^\perp\}| = 5 - |\{i : \pi_i \subseteq [0 : 1]\}| = 5 - 2 = 3.$$

- ① Additive codes
- ② Geometric point of view
- ③ Minimality for additive codes

Minimal linear codes

- **Code:** $\mathcal{C}$ is a linear $[n, k, d]_q$ code
- **Minimal codeword:** $c \in \mathcal{C} \setminus \{0\}$ is minimal if

$$\text{supp}(c') \subseteq \text{supp}(c) \iff c' = \lambda c$$

for some $\lambda \in \mathbf{F}_q^*$

- **Minimal code:** every nonzero codeword is minimal

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Example

Consider $\mathbf{F}_4 = \mathbf{F}_2(\omega)$, with ω a root of $x^2 + x + 1 \in \mathbf{F}_2[x]$.

Let $\mathcal{C}$ be the $[4, 2/2, 3]_2^2$ additive code defined by the following generator matrix

$$G = \begin{pmatrix} 1 & 1 & \omega & \omega \\ 1 & \omega & 0 & 1 \end{pmatrix}.$$

We have that $\mathcal{C} = \{(0, 0, 0, 0), (1, 1, \omega, \omega), (1, \omega, 0, 1), (0, \omega^2, \omega, \omega^2)\}$.

The code $\mathcal{C}$ is minimal.

Proposition (Alfarano, L.M.)

If

$$\binom{nh-2}{k-2}_q \cdot \sum_{i=1}^n \binom{n}{i} (q^h - 1)^i (q^i - q) < \binom{nh}{k}_q,$$

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Corollary (Alfarano, L.M.)

$[n, k/h]_q^h$ minimal additive codes exist whenever

$$n \geq \left\lceil \frac{2k}{h \log_{q^h} \left(\frac{q^{2h}}{q^{h+1} - q + 1} \right)} \right\rceil.$$



Main idea of the proof

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A nonzero codeword $c \in \mathcal{C} \subseteq \mathbf{F}_{q^h}^n$ is not minimal if there exists $c' \in \mathcal{C}$ such that it holds

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Let $B = \left\{ (c, c') \in \mathbf{F}_{q^h}^n \times \mathbf{F}_{q^h}^n : c \neq 0 \text{ and (1) holds} \right\}.$

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An $[n, k/h]_q^h$ additive code is not minimal if it contains the $\mathbf{F}_q$ -space generated by a pair in B .

Strong blocking sets

Definition

An $[n, k, d]_q$ -projective system $\mathcal{P}$ is a **strong blocking set** if for all $H \subseteq \text{PG}(k-1, q)$,

$$\langle H \cap \mathcal{P} \rangle = H.$$

Additive strong blocking sets

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Additive strong blocking sets

Definition (Alfarano, L.M.)

An h - $[n, k, d]_q$ **projective system** $\chi = \{\pi_1, \dots, \pi_n\}$ is an **additive strong blocking set** if

$$\bigcup_{i=1}^n \pi_i$$

is a strong blocking set in $\text{PG}(k-1, q)$.

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Theorem(Alfarano, Borello, Neri)

There is a one-to-one correspondence between (equivalence classes of) minimal codes and (equivalence classes of) strong blocking sets.



Alfarano, Borello, Neri "A geometric characterization of minimal codes and their asymptotic performance", 2020

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Alfarano, Le Meur "Minimal additive codes", 2026

Example

Consider $\mathbf{F}_4 = \mathbf{F}_2(\omega)$, with ω a root of $x^2 + x + 1 \in \mathbf{F}_2[x]$.
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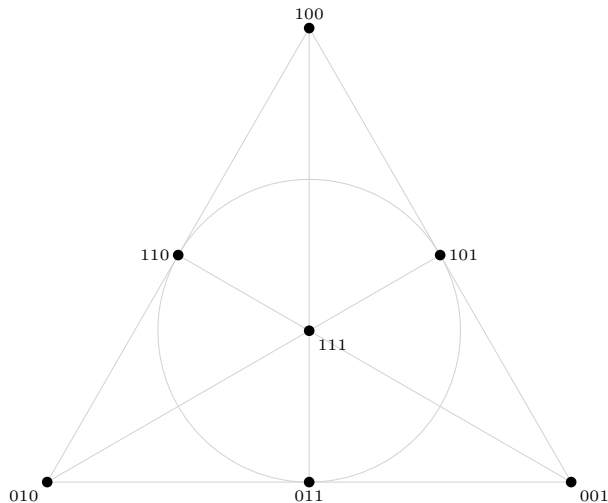
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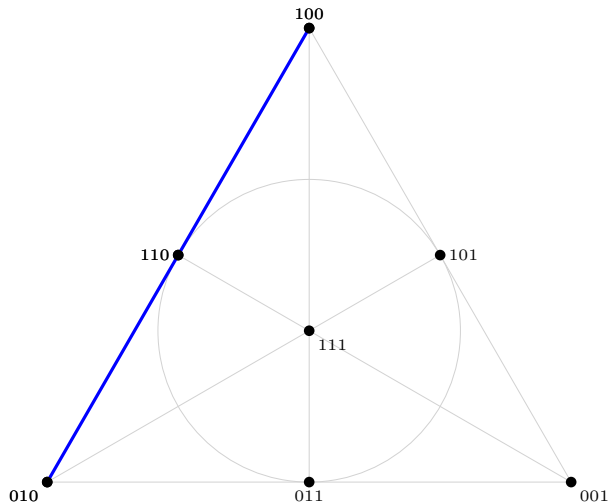
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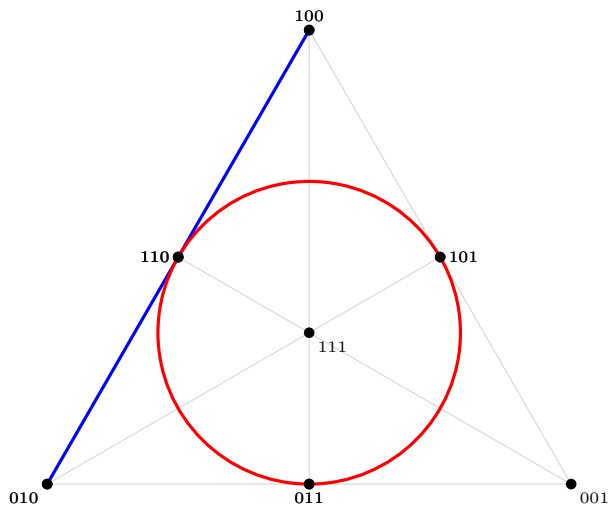
- $\pi_1 = \{[1 : 0 : 0], [0 : 1 : 0], [1 : 1 : 0]\}$



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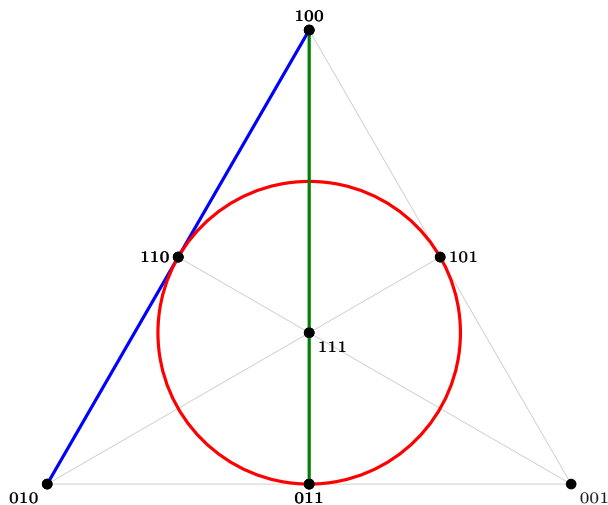
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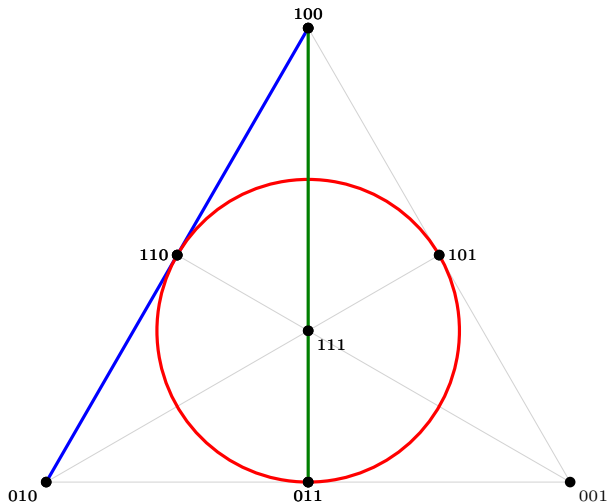
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$\mathcal{M} := \cup_{i=1}^3 \pi_i$ is a strong blocking set, it follows that χ is an additive strong blocking set in $\text{PG}(2, 2)$.

Therefore, the code $\mathcal{C}$ is minimal.



- Determine bounds on the parameters of additive codes
- Determine bounds on

$$m(q, k, h) := \min\{n \in \mathbb{N}_{\geq 1} : \text{there exists a minimal } [n, k/h]_q^h \text{ additive code}\}$$

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Alfarano, Borello, Neri "Outer strong blocking sets", 2024

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Thank you for your attention!

Do you have questions?